Circuit Lab B: hands-on stations (parent guide)

Ten practice stations for the 2027 Division B hands-on part. Set up one station at a time. Give Daniel only his page. Score with the rubric on the parent sheet.

How a real station runs

1. Lay out the parts. Start a timer. A real test has 25–50% of the points in the hands-on part.
2. He may ask how the breadboard rows are joined. Answer that question only.
3. When he raises his hand, test every switch state yourself, then sign.
4. Score: measurements inside the window = full points. Units and sig figs count.

Safety for every station

- Batteries only: 9 V, 3 × AA or 4 × AA. No wall power. No soldering.
- Every LED needs a series resistor. A shorted 9 V battery gets hot: disconnect it while he rewires.
- The meter's mA jack goes in series only. Across a battery it blows the fuse.

Stations

#	Station	Time	Points
1	Multimeter: voltage, current, resistance	15 min	20
2	Mystery resistor: color code and voltage only	15 min	20
3	Build a target resistance	15 min	20
4	Make two LEDs equally bright	15 min	20
5	Power supplied by a source (and a USB label)	15 min	20
6	Troubleshoot a faulty circuit	12 min	20
7	Logic circuit from a truth table	20 min	20
8	Wire a 3-way switch	12 min	20
9	Design a voltage divider	15 min	20
10	Diode and LED: orientation and forward voltage	12 min	20

Windows on the parent sheets assume a 9 V battery between 8.0 and 9.6 V. Measure your battery first: the "× Vs" factors give the exact value for your battery.

Shopping list (one kit for all ten stations)

quantities include spares

Run the stations one at a time and this kit is enough. A resistor assortment kit (1/4 W, E12 values 10 Ω to 1 MΩ) covers every value below. Check that it has 33 kΩ, 47 kΩ and 1 MΩ.

Qty	Item	Stations
3	resistor 330 Ω, 1/4 W	4, 6
3	resistor 470 Ω, 1/4 W	2, 4, 6
3	resistor 560 Ω, 1/4 W	4
3	resistor 680 Ω, 1/4 W	4
5	resistor 1.00 kΩ, 1/4 W	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
3	resistor 2.20 kΩ, 1/4 W	1, 2, 3, 5, 9
3	resistor 3.30 kΩ, 1/4 W	2, 3, 5, 9
3	resistor 4.70 kΩ, 1/4 W	3, 9
3	resistor 10.0 kΩ, 1/4 W	2, 9, 10
3	resistor 33.0 kΩ, 1/4 W	6
3	resistor 47.0 kΩ, 1/4 W	2
4	resistor 1.00 MΩ, 1/4 W	1
2 (one spare)	9 V alkaline battery	1, 2, 4, 5, 8, 9, 10
1	9 V battery snap with leads (stripped or pin ends)	1, 2, 4, 5, 8, 9, 10
1	4 × AA battery holder with leads (6 V)	6
1	3 × AA battery holder with leads (4.5 V)	7
8	AA alkaline cells	6, 7
2	solderless breadboard, 830 holes (a second one lets you leave Station 6 built)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
1 box	jumper wires, male-male (box of pre-cut plus 20 flexible)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
1	basic digital multimeter: DC V, DC mA, Ω, diode test, continuity	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
6	red LED, 5 mm diffused	4, 6, 7, 8, 10
3	green LED, 5 mm	10
3	blue LED, 5 mm	4, 10
4	1N4148 small-signal diode	10
4	SPDT slide switch, 3 pins, 2.54 mm pitch (breadboard fit)	7, 8
2	74HC00 quad NAND, DIP-14	7
2	74HC08 quad AND, DIP-14	7
2	74HC32 quad OR, DIP-14	7
1	masking tape and a marker	2
1	one die (any board game)	6

Multimeter

Any basic digital meter with DC volts, a DC mA range (200 mA), ohms, diode test and continuity beep. It should have about 10 MΩ input resistance on volts (most do; it says so in the manual). Daniel may bring his own “basic” meter to the event if the supervisor allows it, so practice on that one.

Nice to have

- Alligator clip leads (4).
- A small cup or tray for each station’s parts.
- Index cards for the fault cards.

Station 1: parent sheet

Multimeter: voltage, current, resistance

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap with leads; **1** × breadboard; **6** × jumpers; **1** × multimeter; **1** × R1 = 1.0 kΩ (brown black red); **1** × R2 = 2.2 kΩ (red red red); **2** × R3, R4 = 1.0 MΩ (brown black green).

Set up

1. Lay out the parts loose. Do not build the circuit: Daniel builds it.
2. Label the two 1 MΩ parts "1M" with tape.
3. Check the meter: mA jack fuse good (measure a known current first), battery > 8.0 V.
4. Before Daniel starts, measure the real R1, R2, R3, R4 yourself and write them here: R1 ____ R2 ____ R3 ____ R4 ____.

Expected results

Quantity	Nominal (window)	Note
R1, R2 (ohmmeter)	1.00 kΩ (950 Ω to 1.05 kΩ); 2.20 kΩ (2.09 kΩ to 2.31 kΩ)	±5%
Vs	9.00 V (8.00 V to 9.60 V)	battery under a 2.8 mA load
V1	2.81 V (2.33 V to 3.21 V)	× Vs = 0.3125
V2	6.19 V (5.32 V to 6.80 V)	× Vs = 0.6875
I (ammeter and V1/R1)	2.81 mA (2.38 mA to 3.16 mA)	the two methods agree within 2%
V4, ideal meter	4.50 V (3.80 V to 5.04 V)	× Vs = 0.5000
V4, real 10 MΩ meter	4.29 V (3.60 V to 4.82 V)	× Vs = 0.4762; 4.8% below ideal. A 1 MΩ meter reads Vs/3 (0.333 × Vs)
Meter resistance from V4	10.00 MΩ	R_p = R3·V4/(Vs − V4); R_m = R_p·R4/(R4 − R_p)

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
4	R1 and R2 within 5% of your own readings
4	Vs, V1, V2 within the windows, with units and ranges
2	KVL check written (sum within 0.05 V of Vs)
3	ammeter in series, I within the window, % difference under 3%
2	V4 predicted = Vs/2, measured within the window
3	loading explained: the meter (about 10 MΩ) is in parallel with R4
2	meter resistance estimate near the meter spec (10 MΩ on most meters; some cheap meters have 1 MΩ) using measured R3, R4

Safety

- Never put the meter on mA or A across the battery. That is a short. It blows the meter fuse.
- A shorted 9 V battery gets hot. Disconnect it between tasks.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`NV1−0 9 1−2 1k 2−0 2.2k`	V2 (node 2 = top of R2)	`V1 = 9 V  V2 = 6.188 V`	6.188 (matches)
`NV1−0 9 1−2 1k 2−0 2.2k`	current out of the battery	`V1−0 9: 2.812 mA out of +, gives 25.31 mW`	2.812 m (matches)
`NV1−0 9 1−2 1M 2−0 1M 2−0 10M`	V4 with a 10 MΩ meter (the meter is the 10M part)	`V1 = 9 V  V2 = 4.286 V`	4.286 (matches)

Station 2: parent sheet

Mystery resistor: color code and voltage only

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap; **1** × breadboard; **4** × jumpers; **1** × multimeter; **1** × Rk = 1.0 kΩ with a paper tag “Rk = 1.00 kΩ”; **1** × Rx = 3.3 kΩ (orange orange red), bands taped over; **1** × card A = 470 Ω; **1** × card B = 2.2 kΩ; **1** × card C = 10 kΩ; **1** × card D = 47 kΩ; **1** × masking tape.

Set up

1. Measure Rx with your ohmmeter **before** you tape it: Rx(true) = _____. Score Part B against this value.
2. Wrap Rx in tape so no band shows. Mark it “Rx”.
3. Tag Rk with its value. Lay A to D on a card in a row with their letters.
4. Kit resistors may be 4-band (5%) or 5-band (1%). Check the codes below against your parts.

Expected results

Quantity	Nominal (window)	Note
Card codes	<table><tr><th></th><th>value</th><th>4-band</th><th>5-band (1%)</th></tr><tr><td>A</td><td>470 Ω</td><td>yellow violet brown gold</td><td>yellow violet black black brown</td></tr><tr><td>B</td><td>2.20 kΩ</td><td>red red red gold</td><td>red red black brown brown</td></tr><tr><td>C</td><td>10.0 kΩ</td><td>brown black orange gold</td><td>brown black black red brown</td></tr><tr><td>D</td><td>47.0 kΩ</td><td>yellow violet orange gold</td><td>yellow violet black red brown</td></tr></table>	5% or 1% kit
Vk	2.09 V (1.72 V to 2.41 V)	× Vs = 0.2326
Vx	6.91 V (5.99 V to 7.53 V)	× Vs = 0.7674
Rx from the voltages	3.30 kΩ (2.99 kΩ to 3.65 kΩ)	nominal 3.30 kΩ; battery cancels out
Range from Rk ±5%	3.13 kΩ to 3.46 kΩ	Rx × (1 ± 0.05) for his own number
% error	under 2% if the tag shows the MEASURED Rk	vs your ohmmeter Rx; with the printed 1.00 kΩ a ±5% Rk alone gives up to 5%

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
4	Part A: four values right (1 point each), tolerance named
2	schematic: Rk and Rx in series with the battery, meter across each
4	Vk, Vx, Vs with units, Vk + Vx within 0.05 V of Vs
5	Rx within 3% of your true Rx: 5 points; within 6%: 3 points; 3 sig figs
2	tolerance range correct
3	ohmmeter check and percent error; used only voltages in Part B (else 0 for Part B)

Safety

- Low voltage only. Keep the battery off the board while he rewires.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`ma1kv2.093w6.907`	Rx from the two voltages at 9.00 V	`x  Mystery R (Rx)  = 3.3 kohm`	3.300 k (matches)
`cyvnd`	card A from its bands	`470 ohm  +/−5%`	470.0 (matches)
`c3.3k`	bands of the hidden 3.3 kΩ	`4−band: orange orange red gold`	text (matches)

Station 3: parent sheet

Build a target resistance

Parts (exact values)

1 × breadboard; **6** × jumpers; **1** × multimeter; **2** × 1.0 kΩ; **1** × 2.2 kΩ; **1** × 3.3 kΩ; **1** × 4.7 kΩ.

Set up

1. Put the five resistors loose in a cup. No battery at this station.
2. Measure each part yourself first. Your measured networks then score against your parts.

Expected results

Quantity	Nominal (window)	Note
Target 5.50 kΩ	3.3k + 2.2k (series) = 5.50 kΩ (5.22 kΩ to 5.78 kΩ)	build error +0.00%
Target 1.32 kΩ	3.3k || 2.2k (parallel) = 1.32 kΩ (1.25 kΩ to 1.39 kΩ)	build error +0.00%
Target 1.82 kΩ	1k + (4.7k || 1k) = 1.82 kΩ (1.73 kΩ to 1.92 kΩ)	build error -0.02%
Target 2.50 kΩ	(4.7k || 2.2k) + 1k: closest, 0.06% low = 2.50 kΩ (2.37 kΩ to 2.62 kΩ)	build error -0.06%
All 5 in series / all 5 in parallel	12.2 kΩ / 337 Ω	the largest and the smallest

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
3	5.50 kΩ built, measured within 5% (1 point drawing)
3	1.32 kΩ built, measured within 5%
5	1.82 kΩ signed off: 1k + (4.7k || 1k) or another network within 1%
5	2.50 kΩ: (4.7k || 2.2k) + 1k (2.499 kΩ) = 5; any network within 2% = 3; % error right
2	largest 12.2 kΩ, smallest 337 Ω
2	units and sig figs on every value

Safety

- No battery. The ohmmeter works with power off only.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`h1k 1k 2.2k 3.3k 4.7k=1.32k`	top match for 1.32 kΩ	`1.32k (+0%)  3.3k||2.2k`	1.320 k (matches)
`h1k 1k 2.2k 3.3k 4.7k=1.825k`	top match for 1.82 kΩ (1.8246 kΩ)	`1.825k (−0.024%)  1k+(4.7k||1k)`	1.825 k (matches)
`h1k 1k 2.2k 3.3k 4.7k=2.5k`	closest to 2.5 kΩ	`2.499k (−0.058%)  (4.7k||2.2k)+1k`	2.499 k (matches)
`h1k 1k 2.2k 3.3k 4.7k=5.5k`	all 5 in series	`Combos use at most 4 parts. All 5: series 12.2k, parallel`	12.20 k (matches)

Station 4: parent sheet

Make two LEDs equally bright

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap; **1** × breadboard; **6** × jumpers; **1** × multimeter; **2** × red LED, 5 mm; **1** × blue LED, 5 mm; **1** × 330 Ω; **1** × 470 Ω; **1** × 560 Ω; **1** × 680 Ω; **1** × 1.0 kΩ.

Set up

1. Lay out the parts. Do not label the resistors (he reads the bands).
2. Test each LED first with the meter's diode test. A dead LED ruins the station.

Expected results

Quantity	Nominal (window)	Note
Part A best R	(9 – 4.0)/0.010 = 500 Ω → round up: 560 Ω	470 Ω (10.6 mA) also OK
Part A current with 560 Ω	8.93 mA (6.12 mA to 11.3 mA)	V_R = I × 560
Part B red R	(9 – 2.0)/0.010 = 700 Ω → 680 Ω	
Part B red current	10.3 mA (8.12 mA to 12.1 mA)	680 Ω
Part B blue R	(9 – 3.0)/0.010 = 600 Ω → 560 Ω	680 Ω on blue gives 8.8 mA: 17% below red, fails
Part B blue current	10.7 mA (7.99 mA to 12.8 mA)	560 Ω; red/blue differ about 4%
Part C	red takes nearly all 14.9 mA; blue stays dark	current hogging: lower Vf wins

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
4	Part A: series circuit lit, R from (Vs – 2Vf)/I, current within the window
6	Part B signed off: both lit, each with its own R, currents within 15% of each other
4	Part B: V_R measured, I = V_R/R calculated for both, units
3	Part C: blue dark or dim; explains lower Vf takes the current
3	eye sensitivity: the eye sees green-yellow best, so equal current is not equal looks

Safety

- Never connect an LED to the battery without a resistor. It burns out at once.
- Keep the current at or below 20 mA.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`ls9f2c2i10m`	Part A resistor (2 red LEDs, 10 mA)	`r  Resistor  = 500 ohm`	500.0 (matches)
`ls9f2c2i10m`	current with the next E12 value, 560 Ω	`Next std up: E12 560, E24 510  -> I=8.929 mA`	8.929 m (matches)
`ls9f2r680`	red branch with 680 Ω	`i  LED current  = 10.29 mA`	10.29 m (matches)
`ls9f3r560`	blue branch with 560 Ω	`i  LED current  = 10.71 mA`	10.71 m (matches)

Station 5: parent sheet

Power supplied by a source (and a USB label)

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap; **1** × breadboard; **6** × jumpers; **1** × multimeter; **1** × R1 = 1.0 kΩ; **1** × R2 = 2.2 kΩ; **1** × R3 = 3.3 kΩ.

Set up

1. Tag R1, R2 and R3 with their names and values.
2. Optional: give him your own USB charger to read too. Keep it **unplugged**. Work out its levels here first.

Expected results

Quantity	Nominal (window)	Note
R_eq	2.32 kΩ	1k + (2.2k ‖ 3.3k) = 1k + 1.32k
V1	3.88 V (3.25 V to 4.38 V)	× Vs = 0.4310
V23	5.12 V (4.35 V to 5.70 V)	× Vs = 0.5690
I = V1/R1	3.88 mA (3.28 mA to 4.36 mA)	R1 carries all the battery current
Power supplied	34.9 mW (26.3 mW to 41.8 mW)	Vs × I; scales with Vs^2
P1, P2, P3 at 9.00 V	15.0 mW, 11.9 mW, 7.95 mW; sum 34.9 mW	conservation of energy
USB levels	5 V × 3.0 A = 15.0 W, 9 V × 2.22 A = 20.0 W, 12 V × 1.67 A = 20.0 W	use the Output line, not the Input
Charge times	E = 5.000 Ah × 3.7 V = 18.5 Wh; at 9 V: 1.09 h; at 5 V: 1.45 h	t = E/(P × 0.85)

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
3	three voltages with units, V1 + V23 within 0.05 V of Vs
3	I = V1/R1 within the window; reason: R1 is in series with the battery
3	power supplied within the window, in mW with 3 sig figs
3	three part powers; sum within 2% of power supplied
4	USB: 15 W, 20.0 W, 20.0 W (1 point each, 1 for using Output)
4	charge times 1.09 h and 1.45 h (2 each)

Safety

- The USB charger is never plugged in at this station. The rules ban AC in the hands-on part.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`NV1–0 9 1–2 1k 2–0 2.2k 2–0 3.3k`	power supplied at 9.00 V	`V1–0 9: 3.879 mA out of +, gives 34.91 mW`	34.91 m (matches)
`NV1–0 9 1–2 1k 2–0 2.2k 2–0 3.3k`	P1 (last number on the line)	`R1–2 1k: 3.879 V  3.879 mA 1>2 15.05 mW`	15.05 m (matches)
`zc5000v3.7s9a2.22e.85`	charge time at 9 V	`t  Charge time  = 1.0893 h`	1.089 (matches)
`zc5000v3.7s5a3e.85`	charge time at 5 V	`t  Charge time  = 1.451 h`	1.451 (matches)

Station 6: parent sheet

Troubleshoot a faulty circuit

Parts (exact values)

1 × 4 × AA holder with leads (6 V); **4** × AA cells; **1** × breadboard (full size, split rail if you have one); **6** × jumpers; **1** × multimeter; **1** × R1 = 470 Ω; **1** × R2 = 1.0 kΩ; **1** × R3 = 330 Ω; **1** × 33 kΩ (for fault 3); **1** × red LED; **1** × die (or pick a card).

Set up

1. Build the good circuit first (figure). Check that the LED lights. Measure V_A and V_B.
2. Roll the die in secret. Plant that one fault (table). A 6 = your choice.
3. Fault 5 needs a board with split rails: move R2's and the LED's ground to the far half and remove the bridge jumper. On a board with no split, put them on a short loose jumper strip that does not reach the – rail.
4. Hide the die. Write the fault number here: ____.

Expected results

Quantity	Nominal (window)	Note
Good circuit V_A	3.06 V (2.73 V to 3.44 V)	6 V pack, all tolerances
Good circuit LED current	3.20 mA (2.21 mA to 4.40 mA)	visible, not bright
Fault table (6.00 V, Vf 2.0 V)		

#	fault	V_A	V_B	I_LED	sign
0	No fault (normal)	3.06 V	2.00 V	3.20 mA	LED on (dim to medium)
1	LED in backwards	4.08 V	4.08 V	0	LED off; V_B = V_A, so no current in R3
2	Jumper from row A to the – rail (A shorted to ground)	0.00 V	0.00 V	0	LED off; all 6 V across R1; 12.8 mA in R1
3	R3 = 33 kΩ (orange orange orange) in place of 330 Ω	4.06 V	2.00 V	62.5 µA	LED very dim or off; a real LED at this tiny current reads V_B about 1.6–1.8 V (the 2.0 V model says 2.00)
4	Jumper from A to R3 moved one row (open)	4.08 V	0.00 V	0	LED off; V_B about 0 V (B floats)
5	Ground rail split: R2 and LED on the unbridged half	6.00 V	6.00 V	0	LED off; every node reads 6.0 V

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
4	expected V_A = 3.06 V and V_B = 2.0 V written before measuring
4	Vs, V_A, V_B, V_R1 measured with units
5	fault named and placed correctly
3	the reading that proves it (for example V_B = V_A, so R3 has no current)
2	fixed with one change; sign-off
2	open series part: it has the full supply voltage across it

Safety

- The LED in backwards sees about 4.1 V reverse. That is under its 5 V limit. Do not use a 9 V battery here.
- Fault 2 puts 12.8 mA in R1 (77 mW). That is safe for a 1/4 W part.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`NV1–0 6 1–2 470 2–0 1k 2–3 330 D3–0 2`	good circuit: V_A (node 2)	`V1 = 6 V  V2 = 3.057 V`	3.057 (matches)
`NV1–0 6 1–2 470 2–0 1k 2–3 330 D3–0 2`	good circuit: LED current	`D3–0 2: ON  3.204 mA 3>0 6.408 mW`	3.204 m (matches)
`NV1–0 6 1–2 470 2–0 1k 2–3 33k D3–0 2`	fault 3: LED current with 33 kΩ	`D3–0 2: ON  62.47 uA 3>0 124.9 uW`	62.47 u (matches)
`NV1–0 6 1–2 470 2–0 1k`	faults 1 and 4: V_A with no LED current	`V1 = 6 V  V2 = 4.082 V`	4.082 (matches)

Station 7: parent sheet

Logic circuit from a truth table

Parts (exact values)

1 × 3 × AA holder with leads (4.5 V); **3** × AA cells; **1** × breadboard; **20** × jumpers; **1** × multimeter; **1** × 74HC00; **1** × 74HC08; **1** × 74HC32; **3** × SPDT slide switch; **1** × red LED; **1** × 1.0 kΩ.

Set up

1. Do not use the 9 V battery. 74HC chips take 2 to 6 V.
2. Seat each chip across the center gap, notch to the left.
3. Check each switch's common pin with continuity first.

Table B: XOR from four NAND gates (74HC00)

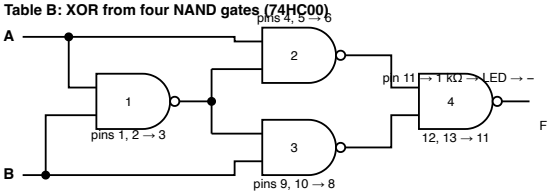
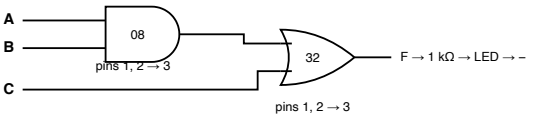


Table A: F = AB + C (74HC08 gate 1 into 74HC32 gate 1)



Each input: SPDT common to the pin; throws to +4.5 V and –. Pin 14 = +, pin 7 = –. Unused inputs to

Expected results

Quantity	Nominal (window)	Note
Table A	F = AB + C. Minterms 1, 3, 5, 6, 7	08 gate 1 (pins 1, 2 → 3) feeds 32 gate 1 (pin 1); C to 32 pin 2; out pin 3
Table B	F = A′B + AB′ = A XOR B	4 NANDs: see figure; output pin 11
Output HIGH	Vs – 0.3 V to Vs (fresh AAs: about 4.6 to 4.9 V)	74HC with about 2.5 mA load; measure Vs first
Output LOW	0.00 to 0.15 V	
LED current	about 2.50 mA	(4.4 – 1.9)/1 kΩ

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
2	Table A expression F = AB + C
6	Table A signed off: all 8 rows right (lose 1 per wrong row)
2	Table B named XOR, expression right
6	Table B signed off with NAND only, all 4 rows
2	drawings with pin numbers, power pins 14 and 7 shown
2	HIGH and LOW voltages measured

Safety

- 4.5 V only. A chip in backwards gets hot: unplug at once.
- Never short an output to + or –.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`t01010111`	Table A output column to an expression	`F = AB + C`	text (matches)
`t0110`	Table B output column	`This is an XOR gate`	text (matches)

Station 8: parent sheet

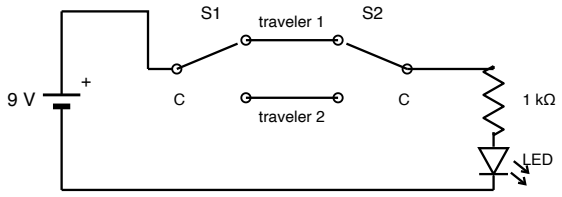
Wire a 3-way switch

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap; **1** × breadboard; **8** × jumpers; **1** × multimeter; **2** × SPDT slide switch; **1** × red LED; **1** × 1.0 kΩ.

Set up

1. Lay out the parts. Find the common pins yourself first, so you can check his answer.



Same traveler = ON. Flip either switch = toggle.

Expected results

Quantity	Nominal (window)	Note
Wiring	S1 common to +9 V; S1 throws to S2 throws (two travelers); S2 common → 1 kΩ → LED → –	figure
Table	same side = on, different = off (XNOR); travelers crossed = XOR	either is a correct 3-way
LED current	7.00 mA (5.52 mA to 8.21 mA)	(Vs – Vf)/1 kΩ
V_R	7.00 V (5.80 V to 7.80 V)	Vs – Vf
Three places	2 SPDT (3-way) + 1 DPDT (4-way) in the middle	

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
3	common pin found by continuity, method explained
3	schematic with two travelers and a series resistor
6	signed off: each switch toggles the LED from all four states
3	table correct (4 rows)
2	V_R and I within the window
3	XNOR / XOR named; three places: 2 SPDT + 1 DPDT

Safety

- The LED needs its 1 kΩ resistor in the single path.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`t1001`	table on / off / off / on	`This is an XNOR gate`	text (matches)
`ls9f2r1k`	LED current with 1 kΩ	`i  LED current  = 7 mA`	7.000 m (matches)

Station 9: parent sheet

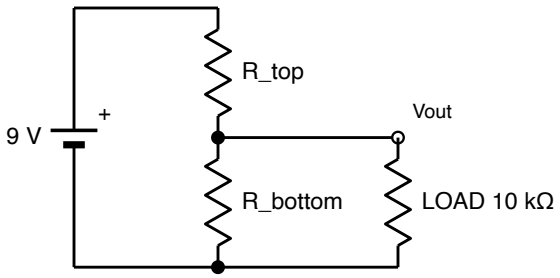
Design a voltage divider

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap; **1** × breadboard; **6** × jumpers; **1** × multimeter; **3** × 1.0 kΩ; **1** × 2.2 kΩ; **1** × 3.3 kΩ; **1** × 4.7 kΩ; **1** × 10 kΩ tagged LOAD.

Set up

1. Tag the 10 kΩ part “LOAD”. Lay out the rest untagged.



Expected results

Quantity	Nominal (window)	Note
Part A, best	top 1k + 1k, bottom 1k: 3.00 V (2.49 V to 3.42 V)	× Vs = 0.3333
Part A, every kit answer within 5%	4.70 kΩ / 2.20 kΩ → 2.87 V; 1.00 kΩ + 3.30 kΩ / 2.20 kΩ → 3.05 V; 2.20 kΩ + 4.70 kΩ / 3.30 kΩ → 2.91 V	nominal, no load
Part B (with 1k+1k / 1k)	2.81 V (2.33 V to 3.21 V)	× Vs = 0.3125: 1k ‖ 10k = 909 Ω; 6% low, fails
Part C, every kit answer within 5% loaded	3.30 kΩ / 1.00 kΩ + 1.00 kΩ → 3.02 V; 4.70 kΩ / 3.30 kΩ → 3.11 V; 4.70 kΩ / 1.00 kΩ + 2.20 kΩ → 3.06 V; 1.00 kΩ + 4.70 kΩ / 1.00 kΩ + 3.30 kΩ → 3.11 V	nominal, 10 kΩ load
Part C example 4.7k / 3.3k	3.11 V (2.59 V to 3.54 V)	3.3k ‖ 10k = 2.48 kΩ
Battery current, 4.7k / 3.3k + load	1.25 mA	

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
3	Part A design from Vout = Vs · R_b/(R_t + R_b), within 5%
3	Part A measured with units, % error
4	Part B: loaded Vout measured; explains the load is in parallel with R_bottom
6	Part C signed off with the LOAD on: Vout / Vs = 0.333 ± 5% (0.317 to 0.350; measure Vs too)
2	Part C calculation shown before building
2	battery current right

Safety

- Small currents only (under 3 mA). No special risk.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`dv9a2kb1k`	Part A: 2 kΩ top, 1 kΩ bottom	`o Vout (across R2) = 3 V`	3.000 (matches)
`dv9a2kb1k/10k`	Part B: same divider with the load	`o Vout (across R2) = 2.812 V`	2.812 (matches)
`dv9a4.7kb3.3k/10k`	Part C: 4.7k / 3.3k with the load	`o Vout (across R2) = 3.11 V`	3.110 (matches)

Station 10: parent sheet

Diode and LED: orientation and forward voltage

Parts (exact values)

1 × 9 V battery; **1** × 9 V snap; **1** × breadboard; **4** × jumpers; **1** × multimeter; **1** × LED P = red; **1** × LED Q = green; **1** × LED R = blue; **1** × diode S = 1N4148; **1** × 1.0 kΩ; **1** × 10 kΩ.

Set up

1. Clear (water-clear) LEDs make the colors hard to guess. Diffused LEDs are fine too: he must still use the meter.
2. Cut the LED legs to the same length so the long-leg rule does not work. Mark which leg is the anode on tape, under the LED.
3. Write down your own diode-test readings: P ____ Q ____ R ____ S ____.

Expected results

Quantity	Nominal (window)	Note
Diode test, forward (about 1 mA)	red 1.6–2.0 V; green 1.8–2.2 V (old type) or 2.4–3.0 V; blue 2.4–3.0 V or OL; 1N4148 0.55–0.75 V	reverse: OL for all
Why blue may read OL	some meters test with under 3 V	then accept OL with the right reasoning
Red LED with 1.0 kΩ	7.00 mA (5.52 mA to 8.21 mA); V_LED 1.8–2.2 V	I = V_R / 1 kΩ
Red LED with 10 kΩ	710 µA (562 µA to 832 µA); V_LED 0.05–0.2 V lower	Vf rises slowly with current
1N4148 forward	V_S 0.62–0.78 V; I = 8.30 mA (6.88 mA to 9.45 mA)	
1N4148 reversed	V_S = Vs (full battery); V_R = 0	no current: the open part has the full voltage

Window = every corner of: resistors ±5%, battery range, LED Vf range (calc.py). A reading inside the window scores full points. Up to twice as far out scores half.

Scoring (20 points)

Pts	Earned for
4	anode found for all four (1 each)
4	diode-test readings with OL the other way; S named silicon
4	1.0 kΩ readings and I within the window
3	10 kΩ readings; says Vf falls a little at low current
3	diode S forward and reversed readings
2	explains reverse: no current, so V_R = 0 and V_S = Vs

Safety

- Do not reverse an LED on 9 V: LEDs are rated for about 5 V reverse. Use the silicon diode for the reverse test.

App check (keys run in the app; screen line shown)

Keys	Gives	Screen line	Python
`ls9f2r1k`	red LED with 1.0 kΩ (Vf 2.0 V)	`i  LED current  = 7 mA`	7.000 m (matches)
`ls9f.7r1k`	1N4148 with 1.0 kΩ (0.7 V)	`i  LED current  = 8.3 mA`	8.300 m (matches)
`NV1–0 9 1–2 10k D2–0 1.9`	red LED with 10 kΩ, Vf 1.9 V	`D2–0 1.9: ON  710 uA 2>0  1.349 mW`	710.0 u (matches)